p-Values:

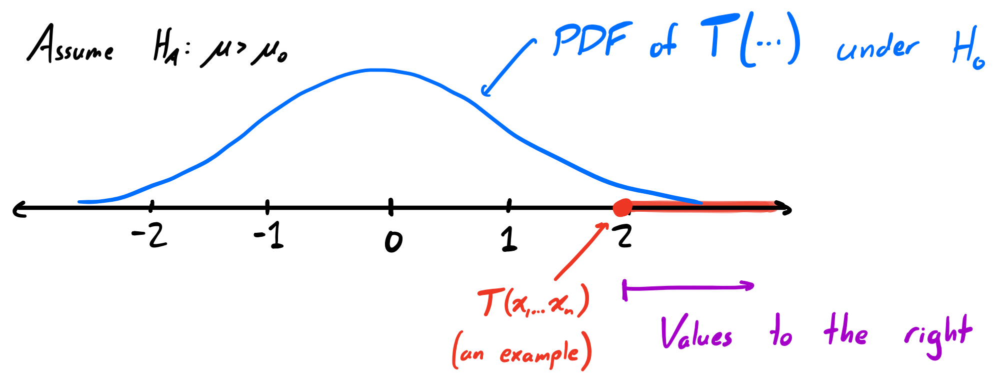
Once we obtain realizations $x_1 \dots x_n$, we will be able to compute:

+ the value of the test statistic $T(x_1, \dots x_n)$
+ the result of the hypothesis test $\mathcal{H}(x_1 \dots x_n)$

this test had some pre-specified size α

After this occurs, we'd like to say how strongly this value $T(x_1 \dots x_n)$ of the test statistic contradicts the null hypothesis.

Recall: Assume $H_A: \mu > \mu_0$



Values to the right of $T(x_1 \dots x_n)$ contradict H_0 in favor of $H_A: \mu > \mu_0$ more strongly than $T(x_1 \dots x_n)$

Def. (p-Value): Given a realization of the test statistic $t = T(x_1 \dots x_n)$, the p-Value associated with t is:

- $\mathbb{P}(T(X_1 \dots X_n) \geq t \mid H_0 \text{ is true})$ if $H_A: \mu > \mu_0$
- $\mathbb{P}(T(X_1 \dots X_n) \leq t \mid H_0 \text{ is true})$ if $H_A: \mu < \mu_0$
- $\mathbb{P}(|T(X_1 \dots X_n)| \geq |t| \mid H_0 \text{ is true})$ if $H_A: \mu \neq \mu_0$

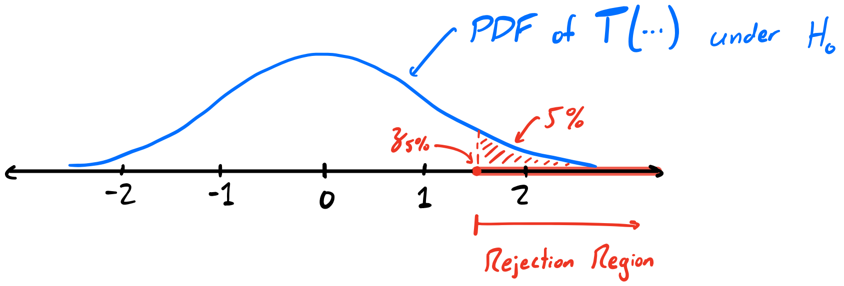
Idea: Smaller p-Value $\rightsquigarrow$ Stronger evidence against H_0

Note: The p-value is a property of the test statistic only, so it depends on the data. Hence, before sampling the p-Value is random. It doesn't look at the rejection region at all. (But H_A does matter)

Note: The p-Value is not the probability that H_0 is true.

Connection to Rejection Region:

Let $H_A: \mu > \mu_0$, $\alpha = 5\%$



Notice: If $T(x_1, \dots x_n) = z_{5\%} \rightsquigarrow$ p-Value = 5%

Moreover:

- $T(x_1, \dots x_n) > z_{5\%} \rightsquigarrow$ p-Value < 5%
- $T(x_1, \dots x_n) < z_{5\%} \rightsquigarrow$ p-Value > 5%

A similar (but not identical) observation holds with different H_A's.

So, we could build a test of size α by rejecting H_0 whenever the p-Value is $\leq \alpha$.

[Exercise: Show this gives a test of size α]

Type II Errors:

So far we have only been concerned with the distribution of our test statistic $T(\dots)$ under the null hypothesis.

$\rightsquigarrow$ This allowed us to study the probability of Type I Error, as

$$\mathbb{P}(T(X_1 \dots X_n) \in R \mid H_0 \text{ is true})$$

But a Type II Error occurs precisely when H_0 is false!

So, to compute $\mathbb{P}(\text{Type II Error}) = \mathbb{P}(T(X_1 \dots X_n) \notin R \mid H_0 \text{ is false})$, we need to know the distribution of $T(\dots)$ when H_0 is false

Ex: $X_1 \dots X_n$ iid, with $n > 30$.

Assume $\mu = \mathbb{E}[X_1]$ is unknown, and $\sigma^2 = Var(X_1)$ is known

$H_0: \mu = \mu_0, \quad H_A: \mu > \mu_0$

Test Stat.: $T(X_1 \dots X_n) = \dfrac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \sim N(0,1)$ under H_0

Rej. Region: $R = [z_\alpha, \infty)$ for a test of size α

Q: If H_0 is false (and $\mu \neq \mu_0$, so that H_A is true), is $T(\dots) \sim N(0,1)$?

A: No! $T(\dots)$ subtracts μ_0 from $\bar{X}_n$, instead of the true mean μ (which is different from μ_0, since H_A is true)

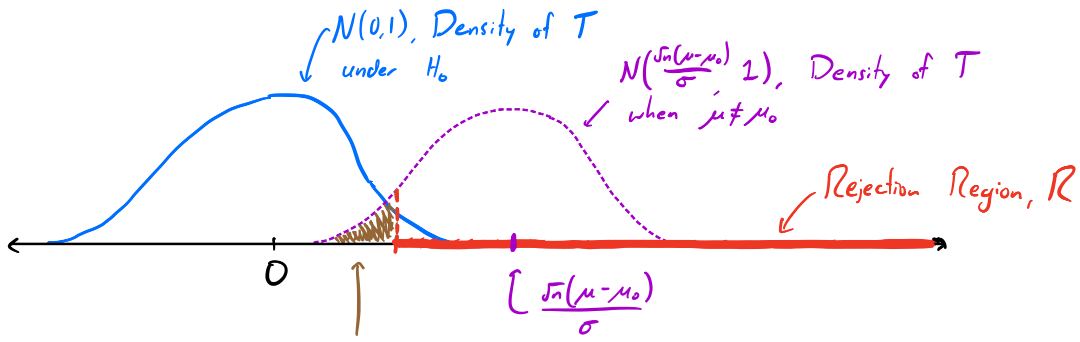
So, if $\mu \neq \mu_0$, what will have a std. normal distribution?

$\rightsquigarrow \dfrac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$ } Not the same as $T(\dots)$ since $\mu \neq \mu_0$

Notice: $T(X_1 \dots X_n) = \dfrac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} = \dfrac{\sqrt{n}(\bar{X}_n - \mu + \mu - \mu_0)}{\sigma} = \underbrace{\dfrac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}_{N(0,1)} + \underbrace{\dfrac{\sqrt{n}(\mu - \mu_0)}{\sigma}}_{\text{A constant}}$

$\therefore T(\dots)$ has a $N\left(\underbrace{\tfrac{\sqrt{n}(\mu-\mu_0)}{\sigma}}, 1\right)$ distribution

Depends on the actual value of μ!



$$\mathbb{P}(T(\ldots) \notin R \mid \mu \neq \mu) = \mathbb{P}(\text{Type II Error})$$

Since the prob. of a Type II Error depends on the true value of the mean, we compute the power of a test as a function of the true parameter value.

$$\text{Power}(\mu) = 1 - \mathbb{P}(\text{Type II Error} \mid \mu = \mu)$$
$$= 1 - \mathbb{P}(\underbrace{T(X_1 \ldots X_n)} \notin R \mid \mu = \mu)$$

Can compute this by knowing that $T(X_1 \ldots X_n) \sim N\left(\frac{\sqrt{n}(\mu-\mu_0)}{\sigma}, 1\right)$
[Exercise]

Small-Sample Hypothesis Tests:

Sometimes, the sample size may not be large enough for the CLT to apply.

$\rightsquigarrow$ A problem! $T(X_1 \ldots X_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}$ or $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{s(X_1 \ldots X_n)}$ may not be approx. $N(0,1)$ under the null $H_0: \mu = \mu_0$.

But, if we know that $X_1 \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$, then we are ok!

+ $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \sim N(0,1)$ for any n, under H_0
+ $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{s(X_1 \ldots X_n)}$ has a t-dist'n with $n-1$ d.o.f., under H_0

$\rightsquigarrow$ We can't use critical values z_α here...

We can now proceed to building rejection regions in the same way as before!

Hypothesis Test w/ One Sample:

Case I: Large Sample Size with Known Variance

- $X_1, \dots X_n$ iid from any distribution w/ unknown μ, known σ^2
- Sample Size: $n > 30$
- Null Hypothesis: $H_0: \mu = \mu_0$
- Test Statistic: $T(X_1 \dots X_n) = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$, approx. $N(0,1)$

H_A:	R (for size α)	p-Value
$H_A: \mu > \mu_0$	$[z_\alpha, \infty)$	$\approx 1 - \Phi(T(x_1 \dots x_n))$ – area right of T
$H_A: \mu < \mu_0$	$(-\infty, -z_\alpha]$	$\approx \Phi(T(x_1 \dots x_n))$ – area left of T
$H_A: \mu \neq \mu_0$	$(-\infty, -z_{\frac{\alpha}{2}}] \cup [z_{\frac{\alpha}{2}}, \infty)$	$\approx 2[1 - \Phi(\lvert T(x_1 \dots x_n)\rvert)]$ – twice the area right of $\lvert T\rvert$

Case II: Large Sample Size with Unknown Variance

- $X_1, \dots X_n$ iid from any distribution w/ unknown μ and σ^2
- Sample Size: $n > 40$
- Null Hypothesis: $H_0: \mu = \mu_0$
- Test Statistic: $T(X_1 \dots X_n) = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$, approx. $N(0,1)$

H_A:	R (for size α)	p-Value
$H_A: \mu > \mu_0$	$[z_\alpha, \infty)$	$\approx 1 - \Phi(T(x_1 \dots x_n))$
$H_A: \mu < \mu_0$	$(-\infty, -z_\alpha]$	$\approx \Phi(T(x_1 \dots x_n))$
$H_A: \mu \neq \mu_0$	$(-\infty, -z_{\frac{\alpha}{2}}] \cup [z_{\frac{\alpha}{2}}, \infty)$	$\approx 2[1 - \Phi(\lvert T(x_1 \dots x_n)\rvert)]$

Case III: Normal Population with Known Variance

- $X_1 \ldots X_n$ iid $N(\mu, \sigma^2)$, with unknown μ, known σ^2
- Sample Size: any n
- Null Hypothesis: $H_0: \mu = \mu_0$
- Test Statistic: $T(X_1 \ldots X_n) = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, exactly $N(0,1)$

H_A:	R (for size α)	p-Value
$H_A: \mu > \mu_0$	$[z_\alpha, \infty)$	$= 1 - \Phi(T(x_1 \ldots x_n))$
$H_A: \mu < \mu_0$	$(-\infty, -z_\alpha]$	$= \Phi(T(x_1 \ldots x_n))$
$H_A: \mu \neq \mu_0$	$(-\infty, -z_{\frac{\alpha}{2}}] \cup [z_{\frac{\alpha}{2}}, \infty)$	$= 2[1 - \Phi(\lvert T(x_1 \ldots x_n)\rvert)]$

Case IV: Normal Population with Unknown Variance

- $X_1 \ldots X_n$ iid $N(\mu, \sigma^2)$, with μ, σ^2 unknown
- Sample Size: any n
- Null Hypothesis: $H_0: \mu = \mu_0$
- Test Statistic: $T(X_1 \ldots X_n) = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, exactly t-dist'n w/ n-1 d.o.f.

H_A:	R (for size α)	p-Value
$H_A: \mu > \mu_0$	$[t_{\alpha, n-1}, \infty)$	$= 1 - T_{n-1}(T(x_1 \ldots x_n))$
$H_A: \mu < \mu_0$	$(-\infty, -t_{\alpha, n-1}]$	$= T_{n-1}(T(x_1 \ldots x_n))$
$H_A: \mu \neq \mu_0$	$(-\infty, -t_{\frac{\alpha}{2}, n-1}] \cup [t_{\frac{\alpha}{2}, n-1}, \infty)$	$= 2[1 - T_{n-1}(\lvert T(x_1 \ldots x_n)\rvert)]$

(where T_{n-1} is the cdf. of the t-dist'n w/ n-1 d.o.f.)

Hypothesis Testing: 2 Samples:

So far, we've tried to make inferences about a single parameter (e.g. mean) in one population.

But many times, we want to compare the same parameter between two populations.

$\rightsquigarrow$ Ex: Randomized Control Trials – split a subset of a population into two random groups. One group recieves a "treatment" but the other doesn't. We may want to see if the two groups have different means.

Setup:

- $X_1, \dots X_n$ are iid with mean μ_X and variance σ_X^2
- $Y_1, \dots Y_m$ are iid with mean μ_Y and variance σ_Y^2

We may have a different number of samples from each pop.

- Assume that X's and Y's are independent
  - $\rightarrow$ "Observing $x_1, \dots x_n$ won't change the dist'n of $Y_1 \dots Y_m$"
- $H_0: \mu_X - \mu_Y = 0$ (usually)

If n, m are large, then by the CLT:

- $\frac{\sqrt{n}(\bar{X}_n - \mu_X)}{\sigma_X} \sim N(0,1)$ (or divided by $s(X_1, \dots X_n)$ for $n > 40$)
- $\frac{\sqrt{m}(\bar{Y}_m - \mu_Y)}{\sigma_Y} \sim N(0,1)$ (or divided by $s(Y_1, \dots Y_m)$ for $m > 40$)

$$\Rightarrow \bar{X}_n \sim N\left(\mu_X, \frac{\sigma_X^2}{n}\right), \quad \bar{Y}_m \sim N\left(\mu_Y, \frac{\sigma_Y^2}{m}\right)$$

Since the X's are independent of the Y's:

$$\bar{X}_n - \bar{Y}_m \sim N\Big(\underbrace{\mu_X - \mu_Y}_{\text{subtract the means}}, \underbrace{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}_{\text{add the variances}}\Big)$$

So:

$$\frac{(\bar{X}_n - \bar{Y}_m) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \sim N(0,1) \; !$$

Putting this all together:

- $T(X_1, \dots X_n, Y_1 \dots Y_m) = \dfrac{\bar{X}_n - \bar{Y}_m}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$ is $N(0,1)$ under $H_0: \mu_X - \mu_Y = 0$,

  if $n, m > 30$ and σ_X^2, σ_Y^2 are known

- $T(X_1, \dots X_n, Y_1 \dots Y_m) = \dfrac{\bar{X}_n - \bar{Y}_m}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$ is $N(0,1)$ under $H_0: \mu_X - \mu_Y = 0$,

  if $n, m > 40$ and $S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)$, $S_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \bar{Y}_m)^2$

$\Rightarrow$ We can use the same rejection regions (in terms of z_α) for a test of size α

with:
- $H_A: \mu_X - \mu_Y > 0$
- $H_A: \mu_X - \mu_Y < 0$
- $H_A: \mu_X - \mu_Y \neq 0$

The only difference is in our construction of the test statistic!

Small-Sample Two-Sample Test:

Same idea! Now, $X_1, \dots X_n \overset{iid}{\sim} N(\mu_X, \sigma_X^2)$ and $Y_1 \dots Y_m \overset{iid}{\sim} N(\mu_Y, \sigma_Y^2)$.

If $H_0: \mu_X - \mu_Y = 0$, then for any n:

$$T(X_1 \dots X_n, Y_1 \dots Y_m) = \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$$ is standard normal under H_0!

$\rightsquigarrow$ So we can do small-sample tests if both pop's are normally distributed and both variances are known!

If σ_X^2, σ_Y^2 are unknown, then

$$T(X_1 \dots X_n, Y_1 \dots Y_m) = \frac{\bar{X}_n - \bar{Y}_m}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$ has a t-dist'n* under H_0!

* the # of d.o.f. is complex — see attached summary notes...

Hypothesis Test w/ Two Samples:

Case I: Large Sample Size w/ Known Variances

- $X_1 \dots X_n$ iid w/ mean μ_X, var. σ_X^2. $Y_1 \dots Y_m$ iid w/ mean μ_Y, var σ_Y^2

  $\to$ Can be any distribution for X's, Y's

- μ_X, μ_Y unknown, but σ_X^2, σ_Y^2 known
- $n > 30$, $m > 30$
- $H_0: \mu_X - \mu_Y = \Delta_0$

$\Rightarrow$ Under H_0, $T(\dots) = \dfrac{(\bar{X}-\bar{Y}) - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$ is approximately $N(0,1)$

H_A:	R (for size α)	p-Value
$H_A: \mu_X - \mu_Y > \Delta_0$	$[z_\alpha, \infty)$	$\approx 1 - \Phi(T(x_1 \dots y_n))$
$H_A: \mu_X - \mu_Y < \Delta_0$	$(-\infty, -z_\alpha]$	$\approx \Phi(T(x_1 \dots y_n))$
$H_A: \mu_X - \mu_Y \neq \Delta_0$	$(-\infty, -z_{\frac{\alpha}{2}}] \cup [z_{\frac{\alpha}{2}}, \infty)$	$\approx 2[1 - \Phi(\lvert T(x_1 \dots y_n)\rvert)]$

Case II: Large Sample Size w/ Unknown Variances

- $X_1 \dots X_n$ iid w/ mean μ_X, var. σ_X^2. $Y_1 \dots Y_m$ iid w/ mean μ_Y, var σ_Y^2

  $\to$ Can be any distribution for X's, Y's

- μ_X, μ_Y unknown, and σ_X^2, σ_Y^2 unknown
- $n > 40$, $m > 40$
- $H_0: \mu_X - \mu_Y = \Delta_0$

$\Rightarrow$ Under H_0, $T(\dots) = \dfrac{(\bar{X}-\bar{Y}) - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{n}}}$ is approximately $N(0,1)$

(H_A, R (for size α), and p-Values same as Case I)

Case III: Normal Populations w/ Known Variances

- $X_1 \dots X_n \overset{iid}{\sim} N(\mu_X, \sigma_X^2)$, $Y_1 \dots Y_m \overset{iid}{\sim} N(\mu_Y, \sigma_Y^2)$
- μ_X, μ_Y unknown, but σ_X^2, σ_Y^2 known
- Any values of n, m
- $H_0: \mu_X - \mu_Y = \Delta_0$

$\Rightarrow$ Under H_0, $T(\dots) = \dfrac{(\bar{X}-\bar{Y}) - \Delta_0}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$ is exactly $N(0,1)$!

(H_A, R (for size α), and p-Values same as Case I)

Case IV Normal Populations w/ Unknown Variances

- $X_1 \dots X_n \overset{iid}{\sim} N(\mu_X, \sigma_X^2)$, $Y_1 \dots Y_m \overset{iid}{\sim} N(\mu_Y, \sigma_Y^2)$
- μ_X, μ_Y unknown, and σ_X^2, σ_Y^2 unknown
- Any values of n, m
- $H_0: \mu_X - \mu_Y = \Delta_0$

$\Rightarrow$ Under H_0, $T(\dots) = \dfrac{(\bar{X}-\bar{Y}) - \Delta_0}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$ is approx. t-distributed w/ ν d.o.f.

# of degrees of freedom equals: $\nu = \dfrac{\left(\frac{S_X^2}{n} + \frac{S_Y^2}{m}\right)^2}{\frac{1}{n-1}\left(\frac{S_X^2}{n}\right)^2 + \frac{1}{m-1}\left(\frac{S_Y^2}{m}\right)^2}$

$\rightsquigarrow$ In practice, you can just round this to the nearest integer

H_A:	R (for size α)	p-Value
$H_A: \mu_X - \mu_Y > \Delta_0$	$[t_{\alpha,\nu}, \infty)$	$= 1 - T_\nu(T(x_1, \dots y_m))$
$H_A: \mu_X - \mu_Y < \Delta_0$	$(-\infty, -t_{\alpha,\nu}]$	$= T_\nu(T(x_1, \dots y_m))$
$H_A: \mu_X - \mu_Y \neq \Delta_0$	$(-\infty, -t_{\frac{\alpha}{2},\nu}] \cup [t_{\frac{\alpha}{2},\nu}, \infty)$	$= 2[1 - T_\nu(\lvert T(x_1, \dots y_m)\rvert)]$

(where T_ν is the cdf. of the t-dist'n w/ ν d.o.f.)

More General Test Statistics: (Whole section is optional! Won't be tested)

So far, we have really only used the normalized sample mean as a test statistic. (In the two-population case, we used the normalized difference b/w sample means)

Still, can we come up with a more general strategy for building test statistics? What if our hypotheses aren't about the mean?

Framework:

- Suppose we have a population distribution with pdf $f_\Theta(x)$, for some unknown Θ.
- Hypotheses - $H_0: \Theta \in \Theta_0$, and $H_A: \Theta \in \Theta_A$, where Θ_0 and Θ_A are mutually exclusive.
- After obtaining a sample, compute:

$$T(x_1, \dots x_n) = \underbrace{\frac{\max_{\Theta \in \Theta_0} \prod_{i=1}^n f_\Theta(x_i)}{\max_{\Theta \in \Theta_A} \prod_{i=1}^n f_\Theta(x_i)}}_{\text{Likelihood Ratio}} \quad \begin{matrix}] \text{ max. likelihood over } \Theta_0 \\ ] \text{ max. likelihood over } \Theta_A \end{matrix}$$

Small T? $\Rightarrow$ Likely data came from Θ_A $\Rightarrow$ Assert H_A is true

Large T? $\Rightarrow$ Likely data came from Θ_0 $\Rightarrow$ Assert H_0 is true

Note: In the above, we don't directly calculate p-Values.

In general, the distribution of $T(\cdot)$ is not "nice"

If you are curious, take more statistics!

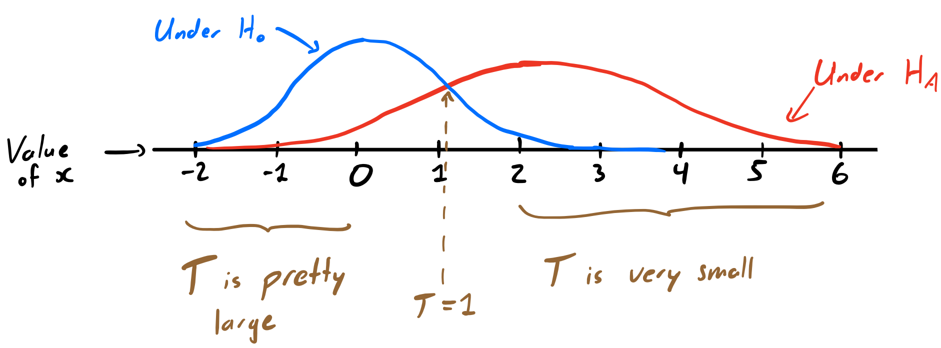
Example:

Assume a simple case where $X \sim N(\mu, \sigma^2)$ for unknown μ, σ^2

Our hypotheses are:

$$H_0: \mu = 0, \sigma^2 = 1 \qquad H_A: \mu = 2, \sigma^2 = 2$$

The possible distributions of X are therefore:



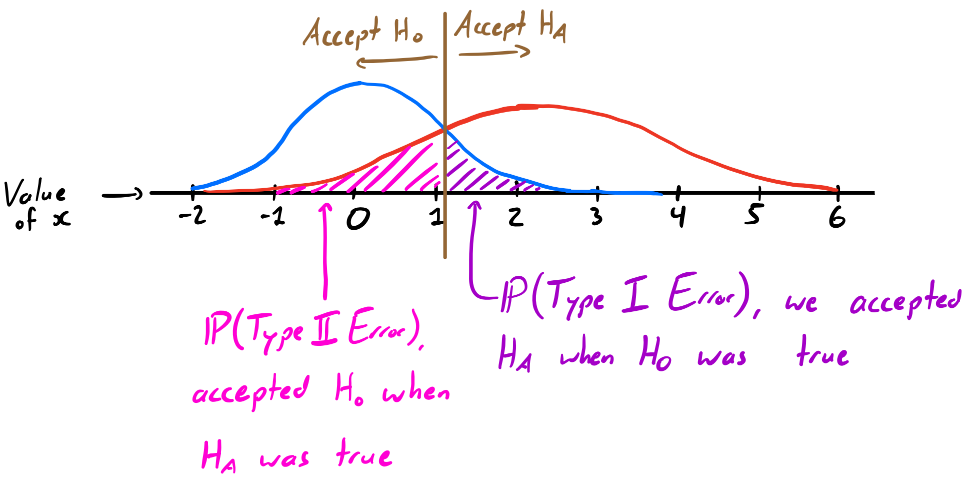
So, a reasonable procedure is to say:

$$\begin{cases} T \geq 1 \Rightarrow \text{Data is more } H_0\text{-likely than } H_A\text{-likely} \Rightarrow \text{Assert } H_0 \text{ is true} \\ T < 1 \Rightarrow \text{Data is less } H_0\text{-likely than } H_A\text{-likely} \Rightarrow \text{Assert } H_A \text{ is true} \end{cases}$$

$\rightsquigarrow$ i.e., $R = [1, \infty)$

Note: We could choose other possible cutoffs for T, this is but one (somewhat natural) choice.

If we do so, how likely are Type I/II Errors?



So clearly, if our hypothesis test looks like:

$$\text{Accept } H_0 \iff T \leq t$$

$$\text{Accept } H_A \iff T > t,$$

for some choice $t > 0$, then as t increases, the prob. of Type I Error decreases, while prob. of Type II Error increases.

$\rightarrow$ A tradeoff!

This approach is extremely common and powerful in statistics!

both in general, and with regards to the power of a test.
